



Time Series Analysis

Seminar 3: Unit roots and ARIMA models



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARIMA: AutoRegressive Integrated Moving Average (model)
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ DF: Dickey–Fuller (test)
- ▣ DW: Durbin–Watson (statistic)
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ GDP: Gross Domestic Product
- ▣ HICP: Harmonised Index of Consumer Prices
- ▣ KPSS: Kwiatkowski–Phillips–Schmidt–Shin (test)
- ▣ MA: Moving Average
- ▣ OLS: Ordinary Least Squares
- ▣ PP: Phillips–Perron (test)
- ▣ SE: Standard Error
- ▣ SES: Simple Exponential Smoothing
- ▣ US: United States
- ▣ ZA: Zivot–Andrews (test)

Today's question and route

- **Question:** does a series have a unit root, how many times must we difference it, and how wide are the forecast intervals of the resulting ARIMA model?
 - ▶ this seminar comes **before** Lecture 3: the section “What you need today” gives every definition the tasks use
- Route
 - ▶ Part A: a Dickey–Fuller statistic by hand; deterministic terms; ARIMA forecasts and intervals; the order of integration of an equation
 - ▶ Part B: unit-root tests on the BET, the S&P 500 and EUR/RON; an ARIMA for Romanian GDP; Romanian inflation with $d = 0$ and $d = 1$
 - ▶ Part C: an open question on structural breaks and an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	a Dickey–Fuller statistic by hand; choosing the deterministic terms	Solved, Proposed	A1
A3, A4	forecasts and intervals of ARIMA(1,1,0) and ARIMA(0,1,1)	Solved, Proposed	A3
A5, A6	the order of integration read from an equation	Solved, Proposed	A5
B1, B2	unit-root tests: BET; S&P 500 and EUR/RON	Solved, Proposed	B1
B3, B4	ARIMA: Romanian real GDP; Romanian inflation	Solved, Proposed	B3
C1, C2	a break or a unit root? what is wrong in an AI answer?	Proposed	B1, B3

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
BET, S&P 500	EODHD	daily, close	2000–2026
EUR/RON	BNR reference rate	daily	2005–2026
Real GDP, Romania	Eurostat (namq_10_gdp)	quarterly, seasonally adjusted	2000–2026
HICP, Romania	Eurostat (prc_hicp_minr)	monthly, 2015 = 100	2004–2026

- Prices and GDP as 100 ln: differences are then log returns or growth rates in %; 12-month inflation $\pi_t = 100(\ln P_t - \ln P_{t-12})$
- In the notebook: `adf_test`, `pp_test`, `kpss_test`, `za_test`, `arima_grid`; no account or key is needed

What you need today (1/4): trends and integration

- **Trend-stationary:** $y_t = \alpha + \beta t + u_t$, u_t stationary: shocks fade, the series returns to the line
 - ▶ **difference-stationary:** $\Delta y_t = \beta + u_t$: shocks are permanent (random walk with drift: u_t white noise)
- $y_t \sim I(d)$, **integrated of order d :** $\Delta^d y_t$ is stationary, $\Delta^{d-1} y_t$ is not
 - ▶ $\Delta y_t = y_t - y_{t-1}$; $\Delta^2 y_t = y_t - 2y_{t-1} + y_{t-2}$
- **Unit root:** the AR polynomial $\phi(z) = 1 - \phi_1 z - \dots - \phi_p z^p$ has the root $z = 1$
 - ▶ then $\phi(z) = (1 - z)\phi^*(z)$ and $\phi^*(L)\Delta y_t = \varepsilon_t$; a stationary AR has all roots with $|z| > 1$ ([Chapter 2](#))
 - ▶ **spurious regression:** regressing one $I(1)$ series on another, unrelated one gives large t and R^2

What you need today (2/4): the Dickey–Fuller test

- ▣ **ADF regression:** $\Delta y_t = c + bt + \gamma y_{t-1} + \sum_{j=1}^k \delta_j \Delta y_{t-j} + \varepsilon_t$, $\gamma = \phi - 1$
 - ▶ H_0 : $\gamma = 0$ (unit root); H_1 : $\gamma < 0$ (stationary); $\tau = \hat{\gamma}/\text{SE}(\hat{\gamma})$; reject if τ is below the critical value
 - ▶ $k = 0$: the Dickey–Fuller (DF) test; k chosen by AIC, at most $12(T/100)^{1/4}$
- ▣ τ does not follow a t distribution: Dickey–Fuller critical values (5%, large T)
 - ▶ no constant -1.94 ; constant -2.86 ; constant and trend -3.41 (Normal: -1.645)
- ▣ Deterministic terms: trend for trending series (log GDP, log prices); constant for series around a level (inflation, returns)
 - ▶ **Phillips–Perron** (PP): no lags, a corrected τ ; same H_0 and critical values

What you need today (3/4): KPSS and breaks

- **KPSS:** H_0 : stationary (around a level or a trend); H_1 : unit root; reject for **large** values
 - ▶ 5% critical values: 0.463 (level), 0.146 (trend)
- **ADF and KPSS together**
 - ▶ ADF rejects, KPSS does not: $I(0)$; ADF does not, KPSS rejects: $I(1)$
 - ▶ neither rejects: inconclusive; both reject: conflict (often a break)
- **Zivot–Andrews (ZA):** ADF with one break at an unknown date; the statistic is the minimum t over all dates
 - ▶ 5% critical values: -4.81 (break in the level), -5.07 (level and trend)

What you need today (4/4): ARIMA models and their forecasts

- ▣ **ARIMA**(p, d, q): $\phi(L)(1-L)^d y_t = c + \theta(L)\varepsilon_t$: an ARMA(p, q) for $\Delta^d y_t$ ([Chapter 2](#))
 - ▶ with $d = 1$, the constant c is the drift: the average change per period
- ▣ **Forecasts**: iterate the model for the levels with future shocks set to 0
 - ▶ error variance at horizon h : $\sigma^2(\psi_0^2 + \dots + \psi_{h-1}^2)$, with $\psi_0 = 1$; 95% interval: $\pm 1.96 \times$ its square root
 - ▶ ARIMA(1,1,0): $\psi_j = 1 + \phi + \dots + \phi^j$; ARIMA(0,1,1): $\psi_j = 1 + \theta$ for $j \geq 1$; random walk: $\psi_j = 1$
- ▣ **Model choice**: AICc ([Chapter 2](#)), lower is better, only between models with the same d
 - ▶ residual check: Ljung–Box with $m - p - q$ degrees of freedom

A1: a Dickey–Fuller statistic [Solved]

Task

- ▣ A monthly interest rate, $T = 120$: OLS gives $\Delta y_t = 0.42 - 0.061 y_{t-1}$, with $\text{SE}(\hat{\gamma}) = 0.025$.
- ▣ 1. Compute τ and $\hat{\phi}$.
- ▣ 2. Compare τ with the 5% Dickey–Fuller value for a regression with a constant, -2.89 , and decide.
- ▣ 3. Say what a standard one-sided t -test would have concluded.
- ▣ 4. Explain why a constant (and no trend) is the right choice for an interest rate.
- ▣ Report: two numbers, a decision and two sentences.

Solution

- ▣ 1. $\tau = -0.061/0.025 = -2.44$; $\hat{\phi} = 1 - 0.061 = 0.939$
- ▣ 2. $-2.44 > -2.89$: do not reject the unit root at 5% ($p = 0.13$); also not at 10% (-2.58)
- ▣ 3. $-2.44 < -1.645$: the t -test would wrongly reject; under H_0 the statistic is not Student
- ▣ 4. An interest rate has no long-run trend; the plausible alternative is "stationary around a mean", which the constant represents.

A2: choosing the deterministic terms [Proposed]

Solution discussed in class

Task

- ▣ The log of a price index ($T = 200$) rises steadily. ADF: $\tau = 1.85$ (no constant), -1.21 (constant), -3.62 (constant and trend); KPSS with trend: 0.09. Model: A1.
- ▣ 1. Say which specification fits the data, and why.
- ▣ 2. Decide at 5% in each specification, with the critical values -1.94 (no constant), -2.88 (constant) and -3.43 (constant and trend).
- ▣ 3. Combine the chosen ADF result with KPSS (5% value 0.146).
- ▣ 4. Say how you would model the series.
- ▣ Report: three decisions, a verdict and one sentence.

A3: ARIMA(1,1,0) forecasts [Solved]

Task

- ▣ $\Delta y_t = 0.6 \Delta y_{t-1} + \varepsilon_t$, $\sigma = 2$; the last two values are $y_{T-1} = 103$ and $y_T = 108$.
- ▣ 1. Compute $\hat{y}_{T+1}$, $\hat{y}_{T+2}$ and $\hat{y}_{T+3}$.
- ▣ 2. Compute ψ_1 and ψ_2 , and the forecast error variances for $h = 1, 2, 3$.
- ▣ 3. Give the 95% intervals.
- ▣ Report: three forecasts and three intervals.

Solution

- ▣ 1. $\Delta y_T = 5$; $\widehat{\Delta y}_{T+1} = 3$, $\hat{y}_{T+1} = 111.00$; $\widehat{\Delta y}_{T+2} = 1.8$, $\hat{y}_{T+2} = 112.80$; $\widehat{\Delta y}_{T+3} = 1.08$, $\hat{y}_{T+3} = 113.88$
- ▣ 2. $\psi_1 = 1.6$, $\psi_2 = 1.96$; variances 4, $4(1 + 2.56) = 14.24$, $4(1 + 2.56 + 3.8416) = 29.61$
- ▣ 3. [107.08, 114.92]; [105.40, 120.20]; [103.22, 124.54]: half-widths 3.92, 7.40, 10.66



A4: ARIMA(0,1,1) and exponential smoothing [Proposed]

Solution discussed in class

Task

- ▣ $\Delta y_t = \varepsilon_t - 0.6 \varepsilon_{t-1}$, $\sigma = 1$; $y_T = 50$ and the last residual is $\hat{\varepsilon}_T = 1.5$. Model: A3.
- ▣ 1. Compute $\hat{y}_{T+1}$ and $\hat{y}_{T+h}$ for $h \geq 2$.
- ▣ 2. Compute the forecast error variances for $h = 1$, $h = 2$ and $h = 5$, and the 95% intervals.
- ▣ 3. Show that the forecast equals simple exponential smoothing ([Chapter 0](#)) and give α .
- ▣ Report: two forecasts, three intervals and α .

A5: the order of integration of an equation [Solved]

Task

- ▣ $y_t = 1.5 y_{t-1} - 0.5 y_{t-2} + \varepsilon_t + 0.4 \varepsilon_{t-1}$.
- ▣ 1. Write the AR polynomial $\phi(z)$ and find its roots.
- ▣ 2. Say whether y_t is stationary, and give d .
- ▣ 3. Write the model as ARIMA(p, d, q) for Δy_t .
- ▣ Report: two roots, d and the ARIMA equation.

Solution

- ▣ 1. $\phi(z) = 1 - 1.5z + 0.5z^2 = (1 - z)(1 - 0.5z)$: roots $z = 1$ and $z = 2$
- ▣ 2. One root on the unit circle: not stationary; the other root is outside: $d = 1$
- ▣ 3. $(1 - 0.5L)\Delta y_t = (1 + 0.4L)\varepsilon_t$: ARIMA(1,1,1),
 $\Delta y_t = 0.5 \Delta y_{t-1} + \varepsilon_t + 0.4 \varepsilon_{t-1}$



A6: integrated or over-differenced? [Proposed]

Solution discussed in class

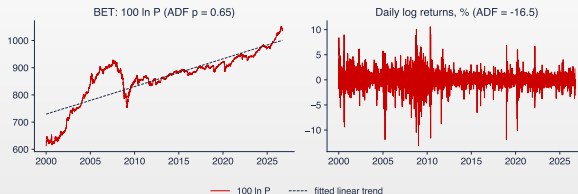
Task

- ▣ Two models: (a) $y_t = 1.8y_{t-1} - 0.8y_{t-2} + \varepsilon_t$; (b) $\Delta y_t = \varepsilon_t - \varepsilon_{t-1}$. Model: A5.
- ▣ 1. For (a), factor $\phi(z)$, give d and write the ARIMA model.
- ▣ 2. For (b), find the MA root and say what y_t really is.
- ▣ 3. Compute $\rho(1)$ of Δy_t in (b) and say how you would detect this case in data.
- ▣ Report: two orders of integration and two sentences.

B1: is the BET log price a random walk? [Solved]

- **Question:** does the BET log price have a unit root, and are its daily returns stationary?
- **Inputs:** BET closes since 2000; $p_t = 100 \ln P_t$ and $r_t = \Delta p_t$
- **Tasks**
 1. Plot p_t with a fitted linear trend, and plot r_t .
 2. Run ADF (lags by AIC), PP and KPSS on p_t with constant and trend.
 3. Run the same tests on r_t with a constant.
 4. Interpretation: if the BET is far below its fitted trend line, should you expect it to return to the line?
- **Report:** a table of six statistics, two verdicts and two sentences
- **Notebook:** section B1

B1: solution [Solved]



	ADF (p, k)	PP	KPSS	verdict
p_t (c, t)	-1.90 (0.653, 19)	-2.35	1.143	I(1)
r_t (c)	-16.50 (< 0.001, 18)	-74.55	0.360	I(0)

- p_t : ADF and PP far above -3.41 , $KPSS > 0.146$: unit root; r_t : both reject strongly, $KPSS < 0.463$: stationary
- Interpretation: no; with a unit root the trend line is not an attractor; at the end of the sample the index is 39 log points from the line, and nothing pulls it back

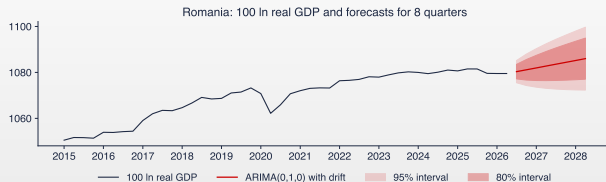
B2: the S&P 500 and EUR/RON [Proposed]

- **Question:** do the S&P 500 and the EUR/RON rate give the same verdicts as the BET?
- **Inputs:** S&P 500 since 2000; EUR/RON (BNR reference rate) since July 2005; both as 100 ln; model: B1
- **Tasks**
 1. Repeat steps 2 and 3 of B1 for both series.
 2. Compare the ADF and PP statistics of the returns, and explain why PP is much more negative.
 3. Interpretation: the EUR/RON rate is managed by the central bank; does the test result say that it is a random walk?
- **Report:** a table (two series, two rows each) and two sentences
- **Notebook:** section B2

B3: an ARIMA model for Romanian GDP [Solved]

- **Question:** which ARIMA model describes Romanian real GDP, and how uncertain is a two-year forecast?
- **Inputs:** real GDP, chain-linked volumes (2010), seasonally adjusted, Eurostat, from 2000Q1; $y_t = 100 \ln Y_t$
- **Tasks**
 1. Choose d with ADF and KPSS on y_t (constant and trend) and on Δy_t (constant).
 2. Fit $\text{ARIMA}(p, 1, q)$ with drift for $p, q \leq 2$ and choose the model with the lowest AICc.
 3. Check the residuals with Ljung–Box $Q^*(8)$.
 4. Forecast 8 quarters with 80% and 95% intervals.
 5. Interpretation: why is the 95% interval after 8 quarters about twice as wide as after 2 quarters?
- **Report:** the tests, the AICc of six models, the chosen model, $Q^*(8)$, the chart and one sentence
- **Notebook:** section B3

B3: solution [Solved]



- ▣ y_t : ADF -2.31 ($p = 0.428$), KPSS $0.198 > 0.146$: $I(1)$; Δy_t : ADF -10.77 , KPSS 0.263 : $I(0)$; so $d = 1$
- ▣ AICc: (0,1,0) 491.9; (1,1,0) 493.8; (0,1,1) 493.7; (1,1,1) 494.9; (2,1,0) 493.7; (0,1,2) 493.6: ARIMA(0,1,0) with drift $\hat{c} = 0.81$ (SE 0.30), $\hat{\sigma} = 2.47$
- ▣ $Q^*(8) = 7.3$ ($p = 0.51$): white-noise residuals; forecast for 2028Q2: 1086.0, i.e. +6.5% in two years, 95% interval [1072.3, 1099.7]
- ▣ Interpretation: for a random walk the error variance is $h\sigma^2$, so the width grows like $\sqrt{h}$: $\sqrt{8/2} = 2$ (half-widths 6.8 and 13.7, ratio 2.00)

B4: Romanian inflation, $d = 0$ or $d = 1$? [Proposed]

- ▣ **Question:** is Romanian 12-month inflation stationary, and how much does the answer change the forecast?
- ▣ **Inputs:** $\pi_t = 100(\ln P_t - \ln P_{t-12})$, HICP, January 2005 to August 2026; model: B3
- ▣ **Tasks**
 1. Run ADF, PP and KPSS on π_t and on $\Delta\pi_t$, with a constant.
 2. Find the best ARIMA($p, 0, q$) with a mean and the best ARIMA($p, 1, q$) by AICc, with $p \leq 3$, $q \leq 2$.
 3. Forecast 12 months with both models, with 95% intervals, and run Ljung–Box $Q^*(24)$ on their residuals.
 4. Interpretation: why do the two models give similar point forecasts but different intervals?
- ▣ **Report:** a table of tests, two models, two forecasts with intervals, two Ljung–Box tests and two sentences
- ▣ Notebook: section B4

C1: a break or a unit root? [Proposed]

- **Question:** do Romanian inflation and GDP have a unit root, or are they stationary around a mean or a trend that shifted once?
- **Inputs:** π_t of B4 and y_t of B3; models: B1, B3
- **Tasks**
 1. Run the Zivot–Andrews test with a break in the level, then in the level and the trend, on both series.
 2. Compare each statistic with its 5% critical value and report the break dates.
 3. Match the dates with events (inflation targeting since 2005, the 2008–2009 crisis, the 2022 energy shock).
 4. Interpretation: why should a break date found by a test be checked against history before it is believed?
- **Report:** a table of four statistics with dates, the chart and a plan for a project
- **Notebook:** section C1

C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant to analyse Romanian GDP and prices. The answer:
- ▣ (a) The ADF p-value for log GDP is 0.43, so GDP is proved to have a unit root.
- ▣ (b) The KPSS statistic for log GDP is 0.198, above 0.146, so KPSS rejects the unit root.
- ▣ (c) Regressing log HICP on the log S&P 500 gives $t = 34.6$ and $R^2 = 0.82$: US stocks drive Romanian prices.
- ▣ (d) GDP growth is stationary, so differencing it once more will make the ARIMA model even better.
- ▣ (e) A random-walk forecast has the same 95% interval width at every horizon.
- ▣ (f) The 5% Dickey-Fuller critical value with a constant is about -2.87, more negative than -1.645, because the Dickey-Fuller distribution is shifted to the left.
- ▣ Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
 - ▶ 2. Report: a list of six verdicts with one line of justification each.

Key takeaways

- ▣ Compare τ with Dickey–Fuller critical values, never with Normal ones; choose the deterministic terms from the plot
- ▣ ADF and KPSS have opposite null hypotheses: read them together
- ▣ Prices, exchange rates and GDP in logs are $I(1)$; returns and growth rates are $I(0)$
- ▣ With $d = 1$, forecast intervals grow with the horizon; the choice of d matters most for long horizons
- ▣ An AI answer is a draft: check the direction of each test and the critical values used

After the seminar

- Lecture 3 develops each topic of today: trends, spurious regression, the Dickey–Fuller distributions, PP, KPSS, breaks, ARIMA forecasting and automatic selection
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: breaks and persistence of inflation in Romania, Hungary, Poland and Czechia
- Reading: [Huang and Petukhina \(2022\)](#), Ch. 4–5; [Hyndman and Athanasopoulos \(2021\)](#), Ch. 9; [Hamilton \(1994\)](#), Ch. 15–17
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

References

- Dickey, D. A., & Fuller, W. A. (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*, 74(366), 427–431. doi:10.1080/01621459.1979.10482531
- Granger, C. W. J., & Newbold, P. (1974). Spurious regressions in econometrics. *Journal of Econometrics*, 2(2), 111–120. doi:10.1016/0304-4076(74)90034-7
- Hamilton, J. D. (1994). *Time Series Analysis*. Princeton University Press. doi:10.2307/j.ctv14jx6sm
- Huang, C., & Petukhina, A. (2022). *Applied Time Series Analysis and Forecasting with Python*. Springer. doi:10.1007/978-3-031-13584-2
- Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and Practice* (3rd ed.). OTexts. otexts.com/fpp3
- Kwiatkowski, D., Phillips, P. C. B., Schmidt, P., & Shin, Y. (1992). Testing the null hypothesis of stationarity against the alternative of a unit root. *Journal of Econometrics*, 54(1–3), 159–178. doi:10.1016/0304-4076(92)90104-Y
- MacKinnon, J. G. (1994). Approximate asymptotic distribution functions for unit-root and cointegration tests. *Journal of Business & Economic Statistics*, 12(2), 167–176. doi:10.1080/07350015.1994.10510005
- MacKinnon, J. G. (2010). Critical values for cointegration tests. Queen's Economics Department Working Paper No. 1227, Queen's University. ideas.repec.org/p/qed/wpaper/1227
- Phillips, P. C. B., & Perron, P. (1988). Testing for a unit root in time series regression. *Biometrika*, 75(2), 335–346. doi:10.1093/biomet/75.2.335
- Zivot, E., & Andrews, D. W. K. (1992). Further evidence on the Great Crash, the oil-price shock, and the unit-root hypothesis. *Journal of Business & Economic Statistics*, 10(3), 251–270. doi:10.1080/07350015.1992.10509904